

SENTIMENT ANALYSIS FOR BUYER & SELLER JOURNEY

A Data Science & Machine Learning Project Report

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Abstract:

The rapid growth of e-commerce has resulted in a massive increase in customer reviews, making it difficult for businesses to analyze feedback manually. This project, "**Sentiment Analysis for Buyer & Seller Journey**," uses **Natural Language Processing (NLP)** and **Machine Learning** to automatically classify customer reviews into **Positive, Neutral, and Negative** sentiments. The dataset was preprocessed by removing missing values, duplicate records, and unnecessary text elements before converting the review text into numerical features using the **TF-IDF (Term Frequency–Inverse Document Frequency)** technique. A **Logistic Regression** model was trained and evaluated to predict customer sentiment. The model achieved an accuracy of **93.94%**, demonstrating effective performance in identifying customer opinions. In addition to sentiment classification, a **Power BI dashboard** was developed to visualize review trends, sentiment distribution, rating patterns, and product insights. The findings of this project can help businesses better understand customer satisfaction, identify areas for improvement, and make informed, data-driven decisions.

Introduction:

Customer reviews have become one of the most valuable sources of information for businesses operating in the e-commerce industry. They reflect customer experiences, product quality, and service satisfaction, helping both buyers make purchasing decisions and sellers improve their products and services. However, manually analyzing thousands of reviews is time-consuming and impractical. Sentiment Analysis, a branch of Natural Language Processing (NLP), enables computers to automatically understand and classify opinions expressed in text. This project applies machine learning techniques to analyze Amazon customer reviews and categorize them into Positive, Neutral, and Negative sentiments. The project covers the complete data science pipeline, including data collection, preprocessing, feature extraction using TF-IDF, model training with Logistic Regression, evaluation, and interactive visualization through Power BI. The generated insights help businesses understand customer behavior, monitor satisfaction levels, and support strategic decision-making.

Problem Statement:

E-commerce platforms receive thousands of customers reviews every day, making it challenging for businesses to manually analyze customer opinions and identify meaningful trends. Important feedback may go unnoticed, affecting customer satisfaction and business growth. There is a need for an automated system that can efficiently process large volumes of textual data, classify customer sentiments, and present insights in an easy-to-understand format. This project aims to develop a machine learning-based sentiment analysis system that accurately categorizes customer reviews into Positive, Neutral, and Negative sentiments while providing interactive visualizations through Power BI to support better business decisions.

Objectives:

The primary objective of this project is to develop a sentiment analysis system that automatically classifies customer reviews into **Positive**, **Neutral**, and **Negative** sentiments using Machine Learning. The project aims to preprocess and clean customer review data, extract meaningful text features using the **TF-IDF** technique, and train a **Logistic Regression** model for sentiment prediction. It also seeks to evaluate the model's performance using metrics such as Accuracy, Precision, Recall, and F1-Score. Additionally, the project aims to create an interactive **Power BI dashboard** to visualize customer sentiments, review trends, product performance, and business insights, enabling organizations to make informed, data-driven decisions.

Dataset Description:

The project uses an **Amazon Customer Reviews** dataset containing customer feedback on various products. The dataset includes information such as **Product Name**, **Brand**, **Review Date**, **Review Rating**, **Review Text**, **Review Title**, **Recommendation Status**, **Helpful Votes**, and **Sentiment**. During preprocessing, missing values and duplicate records were handled, unnecessary columns were removed, and review text was cleaned by eliminating punctuation, special characters, and stop words. The final cleaned dataset consisted of **34,624 customer reviews**, which were used for training and evaluating the sentiment analysis model.

Methodology:

The project follows a structured data science workflow. First, the dataset was collected and cleaned through preprocessing techniques such as handling missing values, removing duplicates, and cleaning review text. The cleaned text was converted into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency)**. The dataset was then divided into training and testing sets using an **80:20 train-test split**. A **Logistic Regression** model was trained on the training data and evaluated on the testing data using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Finally, the results were visualized through an interactive **Power BI dashboard** to present customer sentiment trends and business insights.

Implementation:

The implementation was carried out using **Python** and libraries such as **Pandas**, **NumPy**, **Scikit-learn**, and **Joblib**. Data preprocessing involved cleaning customer reviews and preparing the dataset for machine learning. The cleaned review text was transformed into numerical vectors using the **TF-IDF Vectorizer**, after which a **Logistic Regression** classifier was trained to predict review sentiments. The model achieved an overall accuracy of **93.94%** on the test dataset. The trained model and TF-IDF vectorizer were saved for future predictions. Finally, **Power BI** was used to create an interactive dashboard displaying sentiment distribution, rating trends, top brands, product performance, and recommendation analysis, providing meaningful insights from the customer review data.

Results & Discussion:

The proposed sentiment analysis model was successfully implemented using **TF-IDF Vectorization** and **Logistic Regression**. The dataset was divided into training and testing sets using an **80:20 split**, and the model was evaluated on unseen data. The model achieved an **overall accuracy of 93.94%**, indicating strong performance in classifying customer reviews. It performed exceptionally well in identifying **Positive** reviews, while the performance for **Neutral** and **Negative** reviews was comparatively lower due to the imbalanced nature of the dataset, where most reviews were positive. The evaluation metrics, including **Precision, Recall, F1-Score**, and the **Confusion Matrix**, provided a comprehensive assessment of the model's performance. The results demonstrate that machine learning can effectively automate sentiment classification and provide valuable insights into customer opinions, making it useful for business decision-making.

Power BI Dashboard:

An interactive **Power BI dashboard** was developed to visualize customer review data and sentiment trends. The dashboard includes **KPI cards** displaying Total Reviews, Average Rating, Positive Reviews, Neutral Reviews, and Negative Reviews. It also features visualizations such as Sentiment Distribution, Rating Distribution, Top Brands by Reviews, Top Products, Recommendation Analysis, and Reviews Over Time. Interactive filters for Brand, Rating, Sentiment, and Review Date allow users to explore the data dynamically. The dashboard provides an easy-to-understand representation of customer feedback, helping businesses identify trends, monitor customer satisfaction, and support strategic decision-making.

Conclusion:

This project successfully demonstrated the application of **Natural Language Processing (NLP)** and **Machine Learning** for sentiment analysis of customer reviews. By preprocessing textual data, extracting features using **TF-IDF**, and training a **Logistic Regression** model, the system accurately classified customer sentiments into Positive, Neutral, and Negative categories. The model achieved an accuracy of **93.94%**, highlighting its effectiveness in analyzing customer opinions. The integration of an interactive **Power BI dashboard** further enhanced the project by presenting insights through meaningful visualizations. Overall, the project illustrates a complete end-to-end data science workflow and demonstrates how sentiment analysis can assist businesses in understanding customer feedback and improving products and services.

Future Scope

The project can be enhanced by using larger and more balanced datasets to improve prediction performance for minority sentiment classes. Advanced machine learning and deep learning models such as **Support Vector Machine (SVM)**, **Random Forest**, **LSTM**, and **BERT** can be explored to achieve higher accuracy. Future work may also include multilingual sentiment analysis, aspect-based sentiment analysis, real-time review monitoring, and deployment as a web application or API. These enhancements would make the system more scalable, robust, and suitable for real-world business applications.

References

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